

Computational Intelligence for Optimization

Group Project Statement

2025/26



Contact:

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The challenge

Your group was requested to do a modern interpretation of a well-known painting, Vermeer's *Girl with a pearl earring*. It was decided to do a minimalistic image, and for that purpose, your group will use genetic algorithms to compose an image, as similar to the original as possible, using 100 triangles of possibly different colors, shapes, positions.

How does this famous painting look like, with 100 triangles? That's your quest! With 10 triangles, it looks far from perfect, but still kind of recognizable, right?



Project description

You will use a 300x400 pixels image that is provided with the project.

In addressing the challenge, a lot is being left for each group to decide and define. You will have to define how to characterize each triangle, in terms of size, color and position. Each triangle needs to be of only one color, as in the image above.

Fitness will be defined as pixel-by-pixel Root Mean Squared error between the original image and in the generated one.

Additional challenge

Part of the points for the project are assigned to implementing a challenge. Note that doing more than one challenge will not award you additional points and, if you still decide to do that, you must explicitly inform which challenge does the group want to be evaluated on.

- 1) RGB difference is an intuitive basis for measuring for fitness, however, there are metrics that capture better the way in which the human eye and brain capture and interpret color. Implement a fitness function that uses one of such approaches, then compare results.
- 2) Choose one of the algorithms, other than Genetic Algorithms, that was presented in the theoretical classes and then implemented during practical classes. Generate solutions using that algorithm. Compare the results obtained by that algorithm to the results returned by Genetic Algorithms. The experimental comparison has to be technically and methodologically fair.
- 3) For your GA implementation, consider some of its components (mutation, cross-over, selection) and evaluate their contribution to the overall result.

Dates and deliverables

Project's publication date: 21st of March, 2026

Deadline for group enrollment: 23rd of March, 2026

Deadline for delivery (code+report): 28th of May, 2026, before 00:01.

Submissions must be done on Moodle:

Report must be sent in PDF format and named with the exact group name. As an example, group C01_Beable should submit the report as "C01_Beagle.pdf".

Code must be sent as a zip file, using the same naming convention.

Failing to comply with the deadlines that pertain to the groups (group enrollment and project submission) will result in penalties (-1 point for up to 15 minutes, -2 points for up to 120 minutes, -5 points between 2 and 24 hours). After 24 hours, it will no longer be possible to deliver the project, meaning that the students enrolled in such groups will have a grade of zero in the project.

Should there be any non-trivial discrepancy between the report and the code (eg, mentioning the implementation of a technique that in fact was not implemented) will result in a severe penalization of the project for the whole group.

Evaluation rules

The project must be delivered in groups of 4 students.

Students whose group doesn't deliver a project will get a grade of 0. The same applies to students that do not show up at the date assigned to project defense.

Project defense is strictly mandatory, needs to be done in person at NOVA IMS, and all elements have to be present at the scheduled time.

Grades will be individual, even if the project is a group project. It is therefore more important to work and learn throughout the project than to "lean" on a more experienced colleague.

All students will need to answer questions during the project presentation, about any part of the project – the reasoning, the code, the report.

While it's reasonable for the group to split work, and to use Gen-AI, note that it is totally unacceptable to answer anything in the lines of "I did not work on that part" or "that was ChatGPT/Claude".

You will be evaluated on the code, so make sure that it's properly documented.

You will be evaluated on the report, so make sure your insights and your reasoning are properly presented. Consider that the reader knows everything that was explain in class, so refrain from explaining Genetic Algorithms, etc, on your report. When a decision is made, justify it and mention the alternatives that were considered and why they were discarded. In particular, you probably will want to explain the representation chosen, the fitness function, and any adaptations you made to selection/mutation/crossover.

Finally, you will be evaluated on the presentation. This will supersede the previous evaluations, in the sense that a very good project with a very bad defense will not result in an average grade, but in a very bad grade.

Evaluation criteria

This criteria is tentative, and if needed will benefit from some adjustments.

- 40% Code Functionality: Assessment of the functionality of the code, the correct implementation of the GA techniques and interaction of the various components. While you will work on your projects in a group, you should ensure code is well-structured and works well in conjunction.
- 10% Code Structure: Assessment of the code quality and structure, the code should be clear and well-commented to be inspectable and understandable by anyone.
- 10% Challenge and originality – this corresponds to the implementation of one of the challenges, and to any “wow factor” that you manage to include.
- 40% Report Content: Assessment of the contents of the report. The report should illustrate your project and not contain vast literature reviews (this is a report not an essay). A good report will justify choices undertaken in the project as well as demonstrate findings and results. Reports have a maximum size 10 pages. We anticipate that an excellent or very good report will not need more than 4-5 pages. Use reasonable page/paragraph formatting (all margins 2.5 cm, font size 11, line spacing 1.15). The cover page and appendices do not count, but remember that appendices should not be necessary for us to understand your report.

Keep in mind that we have no tolerance for plagiarism. In NOVA IMS, plagiarism is punished with the failure of both Epoch 1 and Epoch 2 of the exam (failure of the whole course). Reports will also be checked against projects from previous years for plagiarism.

Parting notes

During the practical classes, especially towards the end of the semester, we will have time for project support. This will include a class devoted to going through the project and discussing any questions that may arise.